Latency Decides the Ranking: Effective-Rate Evaluation of RIS Phase Optimization for Mobile Users

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***Abstract*—Phase optimizers for reconfigurable intelligent surfaces (RIS) are usually compared by the rate they reach on the channel they were given. When users move, however, a method delivers its phase to the channel that exists when it finishes. We evaluate methods by this *effective rate*, using measured latencies, in a 3GPP TR 38.901 indoor-office multi-user MISO downlink with a 32-element RIS and pedestrian users (1.5 m/s, 3 GHz, coherence time 28.2 ms). Three results follow. First, the methods with the highest rate at $t = 0$, projected gradient ascent and coordinate descent, need seconds and end below a 0.05 ms single-user heuristic. Second, Adam's effective rate has an interior optimum at $L = 4$ steps (5.980 b/s/Hz); running it to $L = 48$ costs 14.0% although its rate at $t = 0$ rises by 14.8%. Third, a learned (deep-unfolded) update rule beats step-size-tuned Adam at every equal step count, yet on the measured latency axis every learned configuration is dominated by a faster and better Adam configuration, in each of five independent timing rounds. These results assume that the base station refreshes its precoder every frame; without that, the rate at the coherence time would be 11.7–14.2$\times$ lower. All tables and figures are generated by scripts from stored measurements, and we report the measurement traps that produced wrong numbers during the study.**

***Index Terms*—Reconfigurable intelligent surface, mobility, computation latency, effective rate, channel aging, deep unfolding, multi-user MISO.**

I. Introduction

RECONFIGURABLE intelligent surfaces (RIS) reshape the radio channel with a large number of passive, phase-controllable elements [1], [2]. In a multi-user downlink the phases must be optimized jointly with the base-station (BS) precoder, which is a non-convex problem usually attacked with alternating optimization and semidefinite relaxation, block coordinate descent, fractional programming or WMMSE [2]–[6], and, more recently, with learned policies [7].

These methods are almost always compared by the rate they reach on the channel they were given. That comparison is appropriate when the channel is static. When users move, it is not: a method delivers its phase configuration to the channel that exists *when it finishes*, not to the channel it was handed. A slow method answers a question about a channel that is already gone.

This paper makes that observation quantitative for a multi-user MISO downlink assisted by a wall-mounted RIS in a 3GPP TR 38.901 indoor-office scenario [8] with pedestrian users ($v = 1.5$ m/s, $f_c = 3$ GHz, reference coherence time $T_c = 0.423/f_d = 28.2$ ms). Our thesis is that *computation latency is not a secondary metric in mobile RIS systems but a first-class design constraint that decides the ranking of methods*. We support it with measurements, not with a channel-aging formula: every phase is computed on the channel at $t = 0$ and evaluated on a geometrically evolved channel at the method's own measured latency.

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A. Contributions

1) **Effective-rate evaluation.** We define the effective rate of a method as its minimum user rate on the channel that exists at its own completion time (Section IV) and measure it for single-user heuristics, Adam, coordinate descent and projected gradient ascent (PGA). PGA has the highest rate at $t = 0$ (7.117 b/s/Hz) but needs about 5.6 s; its effective rate (4.667 b/s/Hz, an optimistic value of our channel model) falls below that of a 0.05 ms heuristic (5.276 b/s/Hz).
2) **An intrinsic optimum in the step count.** For Adam, the $t = 0$ rate keeps increasing with the number of steps L, but the effective rate peaks at $L = 4$ (5.980 b/s/Hz at 6.65 ms) and then declines. Running to $L = 48$ costs 14.0% of the effective rate although its $t = 0$ rate is 14.8% higher.
3) **Ranking reversal between equal steps and equal milliseconds.** A learned (deep-unfolded) update rule beats a step-size-tuned Adam at every budget $L \in \{1, 2, 4, 8, 16\}$ when both run the same number of steps (win rates 68–84% of test scenarios). Measured on an idle machine in five independent rounds, every one of the five learned configurations is dominated, i.e., some Adam configuration is both faster and better, in all five rounds. One learned step costs 2.7–4.0 Adam steps.
4) **Pricing the modelling assumption.** All effective rates assume that the BS precoder is refreshed from current pilots while only the RIS phase ages. We measure what this assumption is worth: if the precoder aged together with the phase, the rate at T_c would be 11.7–14.2$\times$ lower.
5) **Reproducibility.** Every number in the tables and figures is produced by a script from stored measurement files, and an audit script with 56 assertions checks the reported claims. We report the measurement traps that produced wrong numbers during this work, including a load-contaminated timing that inflated Adam's latency from 17.99 ms to 29.70 ms and flipped a conclusion.

Section II reviews related work, Section III gives the system and channel model, Section IV defines the evaluation framework, Section V the measurement protocol, Section VI the results, and Section VII concludes.

II. Related Work

A. RIS Phase Optimization

Joint precoding and RIS phase design has been addressed with alternating optimization and semidefinite relaxation [2], beamforming under practical phase-dependent amplitude models [3], block-coordinate-descent designs for multicell systems [4], fractional programming for weighted sum-rate maximization [5] and WMMSE-based rate maximization [6]. These methods are evaluated by the objective value they reach; their run time, when discussed, is characterized by complexity order rather than by measured latency relative to the channel's coherence time.

Learning-based designs replace the iterative solver with a trained policy, for example deep reinforcement learning for joint precoding and phase design in multi-user MISO systems [7]. Their usual motivation is speed, yet that speed is rarely related to how fast the channel changes, and the classical solvers they are compared against are rarely tuned or timed under the same conditions.

B. Mobility and Outdated Channel Knowledge

High mobility has been addressed by mounting the surface on the vehicle so that a fast-fading channel becomes slow-fading [9]. The effect of channel aging [10] and of outdated channel state information [11] on RIS-aided systems has been analysed, and predictive beamforming has been proposed to act on a forecast of the future channel [12]. The cost of imperfect channel knowledge at the transmitter is classical in MIMO broadcast channels [13]. In these works the delay between channel acquisition and use is a system parameter. The optimizer's own run time, which differs from method to method and from iteration count to iteration count, is not part of the model.

C. Discrete Phases

Practical surfaces have a few bits of phase resolution [14]. This paper uses continuous phases; the latency dimension of discrete phase design is left to future work (Section VII).

D. Gap Addressed Here

To our knowledge, no prior RIS work (i) evaluates each method on the channel at its own measured completion time, (ii) shows that this creates an interior optimum in the iteration count of a standard optimizer, or (iii) compares a learned update rule with a tuned classical optimizer both at equal steps and at equal measured milliseconds and finds that the two comparisons rank them in opposite order.

III. System and Channel Model

A. Signal Model

A BS with a uniform linear array of $T = 8$ antennas at half-wavelength spacing serves $K = 4$ single-antenna users through a direct link and an RIS of $N = 32$ passive elements (an 8×4 planar array, half-wavelength spacing). With RIS phases $\boldsymbol{\theta} \in [0, 2\pi)^N$, $\mathbf{v} = e^{j\boldsymbol{\theta}}$, and precoder $\mathbf{W} = [\mathbf{w}_1, \ldots, \mathbf{w}_K]$, user k receives

$$y_k = \left(\mathbf{h}_{d,k}^H + \mathbf{g}_k^H \,\mathrm{diag}(\mathbf{v})\mathbf{F}\right) \sum_{j=1}^{K} \mathbf{w}_j s_j + n_k, \qquad (1)$$

where $\mathbf{F} \in \mathbb{C}^{N \times T}$ is the BS–RIS channel, $\mathbf{g}_k^H \in \mathbb{C}^{1 \times N}$ the RIS–user channel, $\mathbf{h}_{d,k}^H \in \mathbb{C}^{1 \times T}$ the direct channel and $n_k \sim \mathcal{CN}(0, \sigma^2)$. Writing $\mathbf{h}_k^H(\boldsymbol{\theta})$ for the bracketed effective channel, the rate of user k is $R_k = \log_2(1 + \mathrm{SINR}_k)$ with

$$\mathrm{SINR}_k = \frac{|\mathbf{h}_k^H \mathbf{w}_k|^2}{\sum_{j \neq k} |\mathbf{h}_k^H \mathbf{w}_j|^2 + \sigma^2}. \qquad (2)$$

For a given phase the precoder is regularized zero forcing on the effective channel, with the MMSE regularizer $\lambda = K\sigma^2/P$ and total power P. The design objective is the minimum user rate,

$$\max_{\boldsymbol{\theta}} \; \min_{k} R_k(\boldsymbol{\theta}), \qquad (3)$$

which gradient methods optimize through the smooth surrogate $-\frac{1}{\beta} \log \sum_k e^{-\beta R_k}$ with $\beta = 8$, evaluated with a log-sum-exp for numerical stability.

B. Geometry and Propagation

The room is $40 \times 20 \times 6$ m. The RIS is mounted on a wall, centred at $(20, 0.1, 3)$ m and facing into the room; the BS is at $(20, 6, 4)$ m, in the panel's boresight and beyond its Fraunhofer distance. Users are dropped uniformly in the room at heights between 1 and 2 m. The carrier is $f_c = 3$ GHz, the bandwidth 20 MHz, the transmit power 20 dBm, the BS antenna gain 5 dBi and the receiver noise figure 7 dB.

All links follow the 3GPP TR 38.901 indoor-office model [8]: its path loss and LoS probability, a Ricean K-factor drawn from its distribution, and log-normal shadowing with Gudmundson spatial correlation (5 m decorrelation distance) [15]. Non-LoS components are built from scattering clusters with explicit geometry. The cascaded BS–RIS–user amplitude follows the two-hop path-loss model of Tang et al. [16] with a $\cos^q$ element pattern. Because an RIS is deployed where the direct path is obstructed, the direct link is attenuated by an additional 40 dB.

C. Mobility

Users walk in a straight line at $v = 1.5$ m/s. Over the observation window the scatterer positions, per-path complex gains, LoS states, Ricean factors, shadowing values and the BS–RIS channel $\mathbf{F}$ are held fixed; only the user positions move, and the channels $\mathbf{g}_k(t)$ and $\mathbf{h}_{d,k}(t)$ are recomputed from geometry at every time instant. Channel variation therefore comes from geometry rather than from a stochastic aging model. As a reference time scale we use the Doppler frequency

$f_d = v f_c/c = 15$ Hz and the coherence time $T_c = 0.423/f_d = 28.2$ ms [17].

D. Known Modelling Limitations

We state three limitations because each one biases the results in a known direction.

- *Frozen scatterers.* Over seconds a walking user would see new scatterers, but the model keeps them. The temporal correlation of the RIS–user channel therefore levels off near 0.5 instead of decaying further, which *understates* the loss of slow methods. We treat latencies up to 200 ms as reliable and also report a stale floor, i.e., a $t = 0$ phase evaluated on an independent channel, as the true asymptote.
- *Independent per-antenna cluster gains.* Cluster gains are drawn independently for each BS antenna instead of being shared across the array. This adds antenna diversity, makes interference nulling easier, and so makes all rates optimistic.
- *Perfect channel knowledge.* Each method is given the exact channel at $t = 0$, and the channel-acquisition delay is not modelled. Imperfect CSI is left to future work.

IV. Latency-Aware Evaluation Framework

A. Effective Rate

Let a method A receive the channel $\mathcal{H}(0) = \{\mathbf{F}, \mathbf{g}_k(0), \mathbf{h}_{d,k}(0)\}$, return the phase $\boldsymbol{\theta}_A$, and need the wall-clock time τ_A to do so.

Definition 1 (Effective rate). The effective rate of A is its minimum user rate on the channel that exists when it finishes:

$$R_A^{\text{eff}} = \min_k R_k\big(\boldsymbol{\theta}_A;\, \mathcal{H}(\tau_A)\big), \tag{4}$$

where the precoder is recomputed from $\mathcal{H}(\tau_A)$.

The precoder is a closed-form digital computation that the BS can refresh every frame from pilots, whereas the RIS phase requires the iterative optimization whose latency is measured here. Definition 1 therefore lets the phase age and keeps the beam fresh. Section VI-D measures how much this assumption is worth.

B. Gain–Aging Decomposition and the Intrinsic Optimum

For an iterative method run for L steps, write $g(L) = \min_k R_k(\boldsymbol{\theta}_L; \mathcal{H}(0))$ for its rate at $t = 0$, $\tau(L)$ for its latency and

$$R^{\text{eff}}(L) = g(L) - \Delta(L), \tag{5}$$

$$\Delta(L) = g(L) - \min_k R_k\big(\boldsymbol{\theta}_L; \mathcal{H}(\tau(L))\big), \tag{6}$$

so that $\Delta(L)$ is the rate lost to aging during the method's own run time. Additional steps raise g with diminishing returns, while Δ grows with $\tau(L)$. An additional step pays off only while

$$g(L+1) - g(L) \;>\; \Delta(L+1) - \Delta(L), \tag{7}$$

and the effective rate peaks at the first L where this inequality fails. We call that point the *intrinsic optimum* $L^\star$. It is a property of the optimizer, the per-step latency and the speed of the channel together, not of the optimizer alone. A slower step or a faster user raises the right-hand side and moves $L^\star$ down. Equation (7) is a bookkeeping identity rather than a theorem: whether R^{eff} is unimodal is an empirical question, which we answer in Section VI-B.

C. Equal Steps versus Equal Milliseconds

Learned update rules are usually compared with classical optimizers at an equal number of steps. If one learned step costs c times as much as one classical step, an equal-step comparison gives the learned method c times the compute budget. We therefore also compare at equal measured time and use Pareto dominance.

Definition 2 (Dominance). Configuration B dominates configuration A if $\tau_B \le \tau_A$ and $g_B \ge g_A$, with one inequality strict.

If B dominates A, there is no latency budget at which A is the better choice. A learned configuration that is dominated by some configuration of the classical optimizer is therefore never worth deploying, regardless of how it compares step for step.

V. Measurement Protocol

A. Scenario Sets

We use two scenario sets, generated with fixed seeds, and never combine numbers from the two in one table.

- *Mobility set* (Tables I and IV, Figs. 1, 2 and 4): 400 scenarios with walking users, seed 11. The aging experiments use 20 scattering clusters per link, close to the 15–19 of the 3GPP indoor-office model; the beam-staleness experiment uses the generator default.
- *Static set* (Tables II and III, Fig. 3): 6000 training, 400 validation and 600 test scenarios. The channel generator is not prefix-stable, i.e., generating n scenarios is not the same as taking the first n of a larger draw, so every checkpoint is evaluated with the sample count it was trained with.

Absolute rates differ between the two sets because their cluster counts differ. Comparisons are always made within one set.

B. Methods

SU-average and *SU-best-of* are closed-form single-user phase alignments (the latter keeps the best of $K+1$ candidates). *Adam* optimizes the soft-min surrogate starting from SU-average. *Coordinate descent* and projected gradient ascent (*PGA*) are run with their own stopping rules; their latencies (about 1.5 s and 5.6 s) were measured on an idle machine. The *learned update* is a deep-unfolded network with L learned steps. Each step receives the gradient of the soft-min rate as input and exchanges information across users, and a separate network is trained for each L without supervision on the rate.

Tuned baseline. At every budget L, Adam's step size is selected on the validation set from ten values in $[0.02, 2.0]$, together with the objective (soft-min or hard min). The selected step size is 0.75 with the soft-min objective at every L, which is an interior point of the grid. Comparing the learned rule against an untuned optimizer would overstate its advantage.

C. Timing

Latencies are wall-clock times on a CPU with one PyTorch thread, measured with `time.perf_counter` for a single scenario, from the shared SU-average starting point to the returned phase. Each configuration is timed 25 times and the median is kept. The learned network gets three warm-up calls. Its input features are precomputed and excluded from the timing, which favours the learned method. The whole measurement is repeated in five independent rounds, and we report the median over rounds. The round-to-round spread is 1.15–1.60× (max/min).

D. Measurement Traps

Three errors occurred during this work. Each one produced a wrong number, and each led to a rule in the protocol.

Timing under load. One timing of Adam with $L = 8$ gave 29.70 ms because a training job was running on the same machine. On the idle machine the same configuration measured 17.99 ms. The slowdown was not neutral: it hit Adam harder than the learned network, and it put a learned configuration on the latency–rate front. Latencies are therefore only measured on an idle machine and only reported with a dominance verdict that holds in every round.

A number without a script. An earlier draft stated that a fresh beam is worth "17×" a stale one. The figure had been computed once, interactively, and could not be regenerated. Measured with a script, the value is 11.7–14.2× (Table IV). Every number in this paper is now produced by a script from stored result files; the tables and figures are generated directly from those files.

Stale derived documents. A human-readable summary of the results had not been regenerated after a latency re-measurement and still listed older timings. The dominance verdict was the same, but the milliseconds were not. Tables are therefore generated from the result files, never copied from summaries.

E. Audit

A separate audit script recomputes the reported claims from the result files and checks 56 assertions; all of them hold. To check that the audit can actually fail, we ran a sabotage test: when the unrolled model's rate was written into the slots of the one-shot models, seven assertions failed.

VI. Results

A. Aging of a Fixed Phase and Effective Rates

Fig. 1 fixes each method's phase at $t = 0$ and tracks the minimum user rate as users walk, with the precoder refreshed at every instant. Rates are flat for the first few milliseconds, drop sharply between roughly 5 and 30 ms, i.e., around T_c, and level off above the stale floor. The more a phase has been optimized, the more it has to lose: by $\tau = T_c$ PGA has fallen from 7.117 to 5.473 b/s/Hz and Adam with $L = 8$ from 6.443 to 5.165, while SU-average has only moved from 4.624 to 4.503.

TABLE I

Rate at $t = 0$, measured latency, and effective rate on the mobility set (400 scenarios, $v = 1.5$ m/s). Stale floor: $t = 0$ phase on an independent channel. †Beyond the 200 ms reliable horizon; optimistic.

Method	t=0 (b/s/Hz)	Latency (ms)	Effective (b/s/Hz)	Stale (b/s/Hz)
SU-average	4.624	0.03	4.624	4.315
SU-best-of	5.276	0.05	5.276	4.336
Adam, $L = 8$	6.443	13.3	5.849	4.326
Adam, $L = 12$	6.631	19.6	5.533	4.370
Adam, $L = 16$	6.754	26.6	5.324	4.381
Adam, $L = 20$	6.835	34.6	5.262	4.364
Coord. descent	6.925	1500.0	4.921†	4.352
PGA	7.117	5600.0	4.667†	4.423

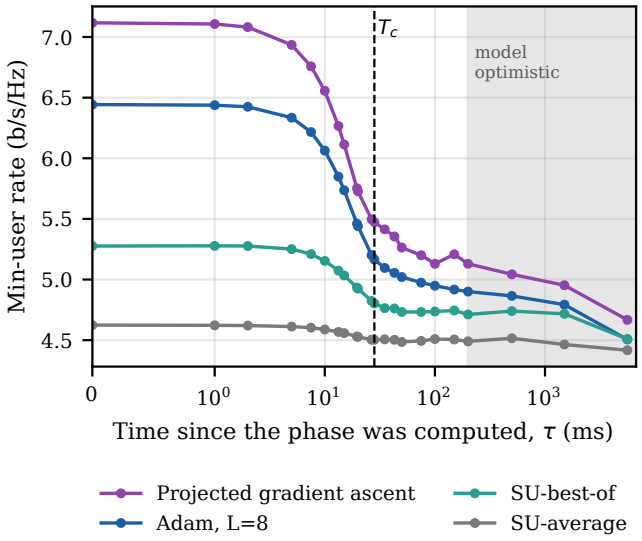


Fig. 1. Minimum user rate of a phase fixed at $t = 0$ as users walk at 1.5 m/s, with the precoder refreshed at every instant. Dashed line: $T_c = 28.2$ ms. Shaded: beyond 200 ms the frozen-scatterer model is optimistic.

Table I evaluates each method at its own measured latency (Definition 1). PGA and coordinate descent have the best $t = 0$ rates but need 5.6 s and 1.5 s, by which time users have moved metres. Their effective rates lie beyond the 200 ms horizon where our model is reliable and are optimistic; the true values lie between the tabulated effective rate and the stale floor. Even the optimistic values are below that of SU-best-of, which takes 0.05 ms. Among the methods in Table I, Adam with $L = 8$ has the highest effective rate (5.849 b/s/Hz); the step sweep below shows that fewer steps do better still.

B. The Intrinsic Optimum

Fig. 2 sweeps the number of Adam steps. The $t = 0$ rate grows monotonically, from 5.276 (SU-best-of, $L = 0$) to 7.037 b/s/Hz at $L = 48$. The effective rate does not: it rises to 5.980 b/s/Hz at $L = 4$ (6.65 ms), stays almost flat at $L = 6$ (5.970), and then falls. At $L = 48$ (79.8 ms) it is 5.145 b/s/Hz, 14.0% below the optimum, although the $t = 0$ rate there is 14.8% higher than at $L = 4$.

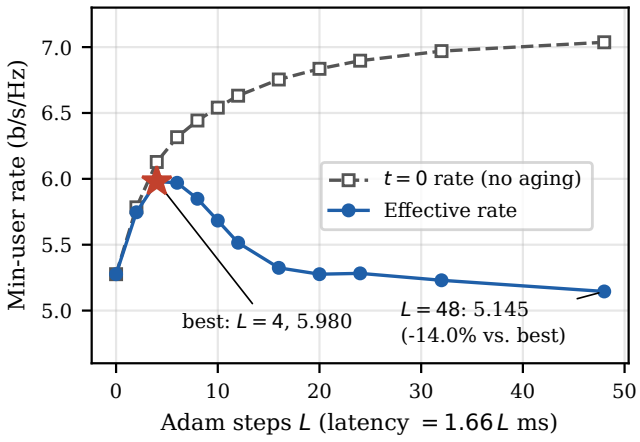


Fig. 2. Adam step sweep on the mobility set. The $t = 0$ rate increases monotonically, while the effective rate (rate at the method's own completion time) peaks at $L = 4$ and is 14.0% lower at $L = 48$.

TABLE II
EQUAL NUMBER OF STEPS, STATIC TEST SET (600 SCENARIOS). ADAM STEP SIZE AND OBJECTIVE SELECTED ON 400 VALIDATION SCENARIOS AT EACH L.

L	Learned (b/s/Hz)	Adam (b/s/Hz)	Ratio	Diff. -2 s.e. (b/s/Hz)	Wins (%)
1	4.460	4.381	1.018	+0.061	74.8
2	4.879	4.693	1.040	+0.160	83.2
4	5.285	5.025	1.052	+0.228	83.7
8	5.593	5.380	1.040	+0.184	83.2
16	5.800	5.739	1.011	+0.039	68.3

In terms of (7): going from $L = 4$ to $L = 6$ adds 0.188 b/s/Hz at $t = 0$, but the additional 3.3 ms of aging cost 0.198 b/s/Hz. Beyond $L = 6$ the effective rate declines; the only exception is a 0.006 b/s/Hz fluctuation between $L = 20$ and $L = 24$. Designing for convergence is therefore not neutral but harmful: the extra steps reduce the effective rate. In this sweep the latency is scaled linearly from the measured Adam-8 latency (13.3 ms, i.e., 1.66 ms per step), which agrees with the per-step cost of 1.60–1.71 ms measured in Section VI-C.

C. Learned versus Classical: Equal Steps and Equal Milliseconds

Table II compares the learned update rule with tuned Adam at equal step counts on the static test set; both start from SU-average (3.539 b/s/Hz). The learned rule is ahead at every budget, the paired difference minus two standard errors is positive at every budget, and it wins on 68–84% of test scenarios. On a step axis, the learned rule is the better optimizer.

Steps, however, are not equally expensive. One learned step takes 4.6–6.5 ms, one Adam step 1.60–1.71 ms, so a learned step costs 2.7–4.0 Adam steps. Table III and Fig. 3(b) place both methods on the measured latency axis. Every learned configuration is dominated by an Adam configuration that is both faster and better, and this holds in each of the five timing rounds separately. For example, the learned rule with $L = 4$ needs 18.41 ms for 5.285 b/s/Hz, while Adam with $L = 8$ needs 13.48 ms for 5.380 b/s/Hz. The ranking from the equal-step comparison reverses completely.

TABLE III
EQUAL MILLISECONDS: EACH LEARNED CONFIGURATION AND THE BEST ADAM CONFIGURATION THAT DOMINATES IT (LOWER MEDIAN LATENCY AND HIGHER RATE). MEDIANS OVER FIVE IDLE ROUNDS; THE DOMINANCE HOLDS IN EVERY ROUND.

Learned update			Dominating Adam		
L	ms	b/s/Hz	L	ms	b/s/Hz
1	5.16	4.460	2	3.20	4.693
2	12.95	4.879	4	6.49	5.025
4	18.41	5.285	8	13.48	5.380
8	36.76	5.593	16	27.31	5.739
16	83.80	5.800	48	78.77	6.107

TABLE IV
FRESH VERSUS STALE PRECODER AT $\tau = T_c = 28.2$ MS, PHASE FIXED AT $t = 0$ IN BOTH (MOBILITY SET, 400 SCENARIOS, $T = 8$).

Phase chosen at $t = 0$	Fresh beam (b/s/Hz)	Stale beam (b/s/Hz)	Ratio
SU-average	3.586	0.306	11.7×
SU-best-of	3.924	0.305	12.9×
Adam, $L = 8$	4.358	0.307	14.2×

Two remarks bound this result. First, the comparison uses $t = 0$ rates on the static set. Aging was not included; since the learned configuration is the slower one in every pair, aging would be expected to widen rather than close the gap, but we have not measured this. Second, the learned timing excludes feature computation, so the latency gap is if anything understated.

D. What the Fresh-Beam Assumption Is Worth

All results above assume that the BS precoder is recomputed from the current channel while only the RIS phase ages (Section IV). Table IV and Fig. 4 measure the alternative in which the precoder ages too. Both curves use the same $t = 0$ phase. With a fresh beam the rate decays moderately (Adam with $L = 8$: from 5.342 to 4.358 b/s/Hz at T_c); with a stale beam it collapses within tens of milliseconds, and at T_c it is 11.7–14.2× lower than with a fresh beam, for every phase choice. The assumption therefore matters, and it must hold in a deployment for the effective rates above to apply: the BS has to refresh its precoder at least once per coherence time. This is the usual frame-by-frame operation of a multi-antenna BS, but it is an assumption of our results rather than a finding.

VII. CONCLUSION

When users move, an RIS phase configuration is worth what it achieves on the channel that exists when it is delivered. Evaluated this way on a 3GPP indoor-office multi-user MISO scenario with pedestrian users, three results follow. First, the methods with the best rate at $t = 0$ (PGA, coordinate descent) take seconds and end up below a 0.05 ms single-user heuristic. Second, the number of optimizer steps has an interior optimum: Adam's effective rate peaks at $L = 4$ steps

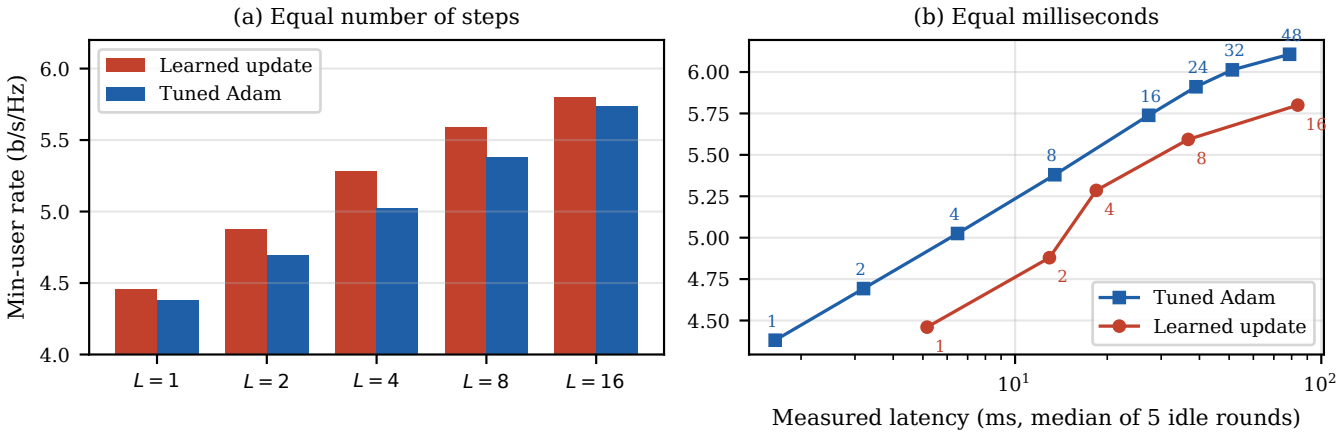


Fig. 3. Learned update rule versus tuned Adam on the static test set. (a) At an equal number of steps, the learned rule is ahead at every budget. (b) On the measured latency axis (numbers: steps), every learned configuration lies below and to the right of the Adam curve, i.e., it is dominated.

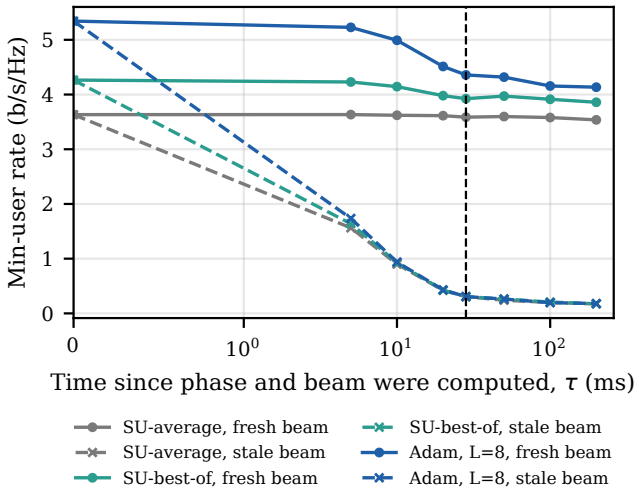


Fig. 4. Same $t = 0$ phase evaluated with a precoder recomputed from the current channel (fresh) or kept from $t = 0$ (stale). Dashed line: T_c.

(5.980 b/s/Hz), and running it to $L = 48$ costs 14.0% even though the $t = 0$ rate keeps rising. Third, a learned update rule that beats a tuned Adam at every equal step count is dominated by Adam at every measured latency, in each of five independent timing rounds. Which method is better depends on whether one counts steps or milliseconds, and in a mobile system only milliseconds matter.

A. Practical Implications

- Stop iterating early. For the pedestrian scenario studied here, a handful of gradient steps (about 7 ms) gives the highest effective rate. The right stopping point moves earlier for faster users or slower steps, by the marginal condition (7).
- Compare methods on measured latency, with the classical baseline's hyper-parameters tuned at each budget, and report dominance rather than equal-step wins.
- Measure latency only on an idle machine, and check that the verdict holds in every repeated round.

B. Limitations and Future Work

The channel model keeps scatterers fixed, so effective rates beyond about 200 ms are optimistic. Cluster gains are drawn independently per BS antenna, which makes all rates optimistic. Channel knowledge is perfect and its acquisition delay is not modelled. The effective-rate results rely on the BS refreshing its precoder at least once per coherence time; without that, rates at T_c are an order of magnitude lower. Three extensions follow directly from these limitations. The first is imperfect CSI, where a learned rule trained on noisy channels might outperform a classical optimizer that overfits the estimate; clean-trained and noise-trained models must be reported separately. The second is discrete phases (1–3 bits), including the latency of discrete optimizers. The third is faster users, for which the intrinsic optimum should move towards fewer steps.

Acknowledgment

[Acknowledgment text, e.g., advisor and computing resources, or remove this section.]

Data and Code Availability

The scripts that generate every table and figure, and the stored measurement files they read, are available at [public repository URL or "upon request"].

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